Agentic Host Environment

Agent LLM / Host Orchestrator (Claude / Cursor / Roo Code)

1. Tool Policy & Parameter Hints

2. Trajectory Ingest (Tool, Params, Output)

3. Temporal & Causal Retrieval

4. Macro Skill Matching & Expansion

Macro Skills & Policy Engine

Macro Distillation Miner - 10-step Window Clustering - Slot Parameterization - Hierarchical L2/L3 Skills

Tool Recommendation Matrix - Laplace-smoothed Success Rate - Bayesian Beta Confidence - Error-Pattern Auto-Correction Hints

Model Context Protocol (MCP) Boundary

MATP Unified MCP Server (JSON-RPC stdio)

Ingestion & Distillation Pipeline

Background Abstraction Engine (Fact Extraction & Token Compression)

Deterministic Local Vectorizer (Subword n-gram hashing + TF saturation)

Resource Canonicalizer (db:*, file:*, api:*, git:*)

Enqueue Enrichment

Asynchronous Enrichment

SQLite Job Queue

Async Neural Worker (Ollama / Qwen2.5 Local)

Update Neural Summary

Causal Reasoning & Temporal Retrieval

Temporal Decay & Clock - Logical Discrete Clock (t_curr) - Exponential Decay exp(-lambda * dt) - Per-Task Adaptive Lambda Tuner

Cross-Session Causal DAG Engine - Writer Supersession (W = 0) - Reader Staleness (W = 0.5 × taint) - Transitive Suspect Propagation

Hybrid Scorer R(q, v) = (alpha*Dense + (1-alpha)*Sparse) × W_DAG × exp(-lambda * dt)

Durable Multi-Domain Storage (SQLite)

MATP Persistent SQLite Buffer - Trajectories & Tuples - Resource Versions & Accesses - Causal DAG Edges - Macro Skills & Slots - Learned Parameter Corrections - Durable Job Queue